

INSTRUCTIONAL INNOVATION: Experiential Learning

*Experiential teaching methods can foster deep learning and result in durable and flexible knowledge, says Professor of Finance and Founding Academic Director of the Financial Research and Trading Lab, **Tom McCurdy**. In this article, he and Professor **Rosamund Woodhouse** — Director of the Office of Health Sciences Education at Queen's University — describe how innovative instructional facilities can be used to enhance the educational experience.*

If creativity and innovation are crucial in the world of business, then how can business education best prepare future managers and leaders to become innovative? Research on student learning suggests that certain types of instructional innovation will promote solutions to this educational challenge. This article will focus on how experiential teaching methods promote deep as opposed to surface learning.

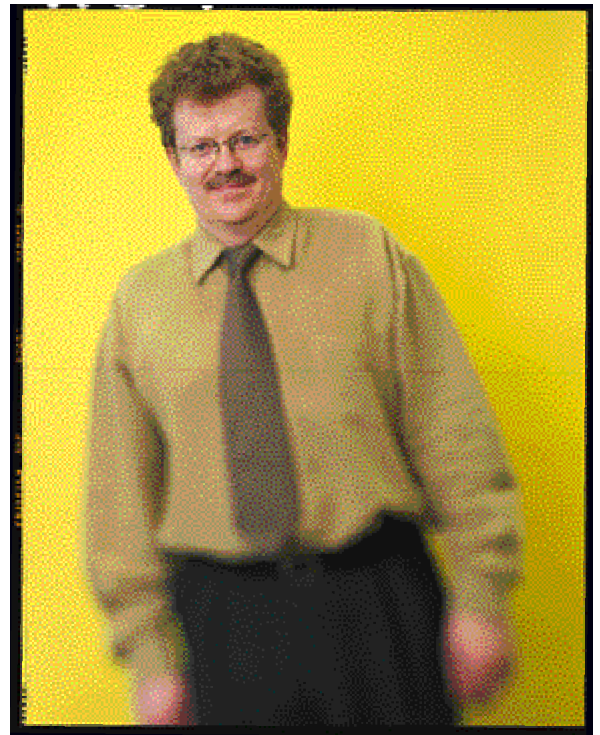
One of the most important variables influencing learning outcomes is the student's approach to learning. A significant body of research has demonstrated that university students hold varying conceptions about knowledge and how it is acquired, and that this influences their learning. Students with a *deep* approach to learning are interested in the meaning and interrelationships of the ideas about which they are learning. Such students learn purposefully, by connecting ideas and questioning them in the light of other information or experiences. Significantly, they are intrinsically

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Professor Tom McCurdy

motivated to learn, often going beyond what is prescribed for their courses as they explore concepts and their implications. Subsequent research has shown that that by approaching learning in this way, deep learners tend to achieve better understanding of material, and to remember it better, than students who take a *surface* approach to learning.

Students with surface approaches to learning tend to restrict their study to the materials *absolutely required* for the course. They overuse memorization as a study strategy, with the goal of being able to reproduce material — sometimes with very little concern about whether



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or not they understand it. These surface learning strategies provide a very poor foundation for subsequent use of concepts or skills, let alone innovative approaches to new problems.

As educators, how can we promote deep learning? Evidence shows that students' approaches to learning can be adapted. Instructional approaches and assessments that actively involve learners in *working on* the material are critical. Activities that foster deep learning require students to clarify their understanding of ideas, to relate them to what they already know, to look for patterns and meanings, and to appraise evidence and the conclusions drawn from it. The likelihood that students will adopt deep learning strategies is also dependent on whether assessments require these same kinds of activities, on whether the overall workload allows time for students to adequately engage in these tasks, and on whether students have opportunities for feedback on their learning while it occurs (instead of just at the end of the course). While this can be achieved to some extent using traditional 'paper and pencil' methods (such as cases, problem sets and assignments), in order to prepare our students to work in professional environments, it is useful to go further.

In practice, professionals are rarely provided with neatly-defined problems and all required sources of information, or with cues about the appropriate problem-solving technique to use. They need to be able to use their knowledge and skills in complex, real-time environments, and to develop skills in finding appropriate data and tools. In many professional education programs, this ability is developed through supervised 'hands-on' experience in real-world settings. However, there are few such 'clinics' or 'teaching practice' settings to enable our learners to obtain such experience. Hence, the value of a learning facility that enables students to apply, critically examine and develop what they have learned in the classroom, using the kinds of data and tools that they would find in the workplace, but in a 'safe' and controlled environment.

The Financial Research and Trading Lab (FRTL) at the Rotman School provides such an environment. It enables students to engage in deep learning by applying and extend-

ing the knowledge and skills introduced in traditional classroom settings. For example, students in the Lab use real-time information from actual markets and for the whole range of financial securities and companies found in the workplace. Working with actual data, as well as using and developing financial tools, enhances their understanding of why and how financial securities are traded, how they are valued, sources of risk and return, etc. Because working in the Lab enables immediate feedback on their actions, it encourages students to test ideas and learn in an individualized and meaningful way. As long as the learning experience is designed to go beyond rote repetition of highly structured exercises, this practical experience fulfills key conditions for deep learning.

In addition to enabling students to practice applying what they learn in the classroom, the FRTL offers the opportunity for students to learn *more*. By working with real data and industry-standard tools, they are engaged in a cycle of experiential learning. The Lab enables students to try out strategies and procedures they have learned about, and to experience the consequences of their actions. As they confront theories and analytical tools with data, they encounter new and unanticipated issues or problems. This feedback can stimulate students to ask new questions, and hence lead to new ways of understanding the concepts and problems that they are learning about. These in turn can be tested, perpetuating the learning cycle. Therefore, in addition to providing invaluable practical experience, experiential learning can foster the spirit of inquiry that underlies innovation.

The challenge for business schools is to develop educational approaches that foster deep understanding of issues and approaches, and provide incentives to challenge and explore new possibilities. Experiential teaching methods can foster deep learning, resulting in durable and flexible knowledge and the intrinsic motivation for life-long learning.

These outcomes are essential for our graduates if they are to fulfill their future roles as innovative leaders. The use of experiential teaching methods in the Financial Research and Trading Lab is just one example of how the Rotman School is innovating in the realm of business education. **RM**

Creative Idea Generation: Beyond Brainstorming

By Peter A. Heslin

Generating viable ideas for new products, services, and processes is imperative for most organizations to survive in our dynamic era of emerging technologies and hyper-competition. Although brainstorming remains a widely-used approach for generating ideas, its limitations have led to the development of alternate approaches, one of which is called **brainwriting**.

Brainwriting entails four people sharing written ideas as they are generated. Participants are seated at a table where they write an idea on one of about 25 small pieces of paper, before passing it to the person seated to their right. Accountability is increased and 'social loafing' — a low level of contribution that results from people feeling that they are not personally accountable — is reduced by each person writing in a different color pen, which makes it clear who produced each idea.

As participants receive a slip from the person to their left, they are instructed to read the ideas on it, add one of their own, and pass it on. If participants cannot come up with an idea in a reasonable period of time, they can pass the slip on without writing anything. When each participant receives a completed slip with four ideas on it, they are instructed to read those ideas before placing the slip in the centre of the table.

The second stage is a 15-minute recall period where individuals are re-seated in different corners of the room and asked to individually list as many ideas as possible from the first stage, including their own. This stage is intended to encourage careful attention to the ideas generated by other group members. In the third stage, participants remain apart and are asked to continue generating ideas individually (without sharing) for an additional 15 minutes.

Research indicates that brainwriting groups generate, on average, more unique ideas than brainstorming groups. Brainwriting minimizes the 'production blocking' that occurs during brainstorming, whereby the inability to immediately share one's ideas leads to them being either forgotten or self-evaluated as not 'good enough' to share. It encourages the careful processing of other group members' ideas, fosters competition, and discourages social loafing. Thus, it is not surprising that the brainwriting procedure appears to be more effective than brainstorming at stimulating the production of creative ideas.

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